

Temporal data mining and application to anesthesia monitoring

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https://github.com/andrew-m-kehl/hypo_pred

Abstract

The objective of this research project is to develop predictors of hypotension by identifying and characterizing critical points of the arterial blood pressure wave form. To achieve this, we investigate the use of Signal Temporal Logic and StEval, a tool developed by Verimag, for physiological signal analysis. Because StEval has fixed temporal bounds, this study uses an algorithm to scale the temporal evaluation windows of STL formulas which are adjusted proportionally to a patient's heart rate. These scaled formulas are used to identify key waveform characteristics including: diastolic minima, systolic peaks and the dicrotic notch. The extracted features are then used to construct an early warning system capable of predicting hypotensive events before onset. In addition, this study investigated the utility of the dicrotic notch within the STL framework. While preliminary studies suggest that the dicrotic notch may contain clinically relevant information, further investigation is required to assess its reliability and predictive value for hypotension. Overall, the results demonstrate that scaled signal temporal logic is an effective approach for real-time arterial blood pressure characterization and shows promise for hypotension prediction.

1 Introduction

1.1 Research Problem

During surgery, Intraoperative Hypotension (IOH) has been increasingly associated with adverse postoperative and long-term patient outcomes. Inadequate monitoring or delayed detection of low blood pressure can lead to serious complications [Dmania, 2019]. Consequently, there is growing interest in developing methods that can identify patients at risk of hypotension before a critical event occurs.

1.2 Background

Intraoperative Hypotension

The Anesthesia Patient Safety Foundation [Yerdon *et al.*, 2024] defines IOH as an event that occurs when the Mean Arterial blood Pressure (MAP) falls under 65 mmHg for 1

minute. Although the exact definition of IOH varies, it can be summarized when a patient's blood pressure goes below a safety threshold for an amount of time in which critical organs do not receive enough blood. According to a study from the Department of Anesthesiology, Belgium [R. Vossen, 2022] and research from the Anesthesia Patient Safety Foundation [Yerdon *et al.*, 2024], IOH has been linked to a higher risk of acute kidney injury, myocardial injury (bruised heart muscles), cerebral ischemia (low blood flow to the brain), postoperative delirium (disturbance of attention, awareness, and cognition) and even postoperative mortality (death) in patients undergoing non-cardiac surgery under general anesthesia.

State of the Art

To monitor blood pressure during surgery, clinicians currently use either an external sensor which is placed on the finger, or a catheter sensor which is inserted within the artery [Bergholz *et al.*, 2023]. The measurements from the sensor will trigger an alarm if the vital signs of the patient fall outside of a critical range. At this point the medical team reacts using their best professional expertise and judgement. However, within this window of when vital signs cross critical thresholds to when the medical team reacts, can be critical to the patient's health [Dmania, 2019]. Delays in recognition and reaction to hypotensive events can increase the risk of adverse effects. Consequently, there is growing interest by medical professionals in developing predictive methods that can identify these conditions in advance, enabling early intervention and improved patient outcomes.

Related Work

Medical Publications from Critical Care [Yoon *et al.*, 2020] and Medicina [Shim *et al.*, 2025] show that hypotension might be predictable using Machine Learning Models (MLM). Yoon *et al.* [2020] used MLMs to analyze trends and build risk score trajectories. The risk score trajectories were used to compute the probability of a hypotensive event and tended to rise as the MAP of the patients got closer to hypotension. Shim *et al.* [2025] used MLMs to transform input data into abstract and complex representations and create binary class predictions. A study released in the IEEE journal of Biomedical and Health Informatics [Aubouin *et al.*, 2025] proposed a comprehensive machine learning approach using a forward sliding window framework to extract MAP trends,

signal variability and regression slopes to predict hypotension. According to studies from the Department of Anaesthesiology and Critical Care, Lariboisiere Hospital in Paris [Coutrot *et al.*, 2019], a combination of the relative variations in diastolic notch height and perfusion index showed a high predictive power of intra-operative hypotension.

1.3 Objective and Hypothesis

While machine learning techniques have been increasingly used to predict events such as hypotension, they often rely on complex and opaque features that are difficult to interpret and verify. In contrast, this research project aims to apply a more formal approach based on signal temporal logic which can explicitly encode constraints and temporal relationships that are not directly observable in the data. Building upon the research established by Coutrot *et al.* [2019] and Aubouin [2025], the objective of this research project is to develop predictors of hypotension by identifying and characterizing critical points of the arterial blood pressure waveform. To achieve this, key characteristics of the arterial blood pressure waveform are represented as formulas in Signal Temporal Logic (STL), and evaluated over sliding time windows for satisfaction of the formula condition. These key characteristics including: the diastolic minimum, systolic peak, and diastolic notch, can be used to make calculations on MAP trajectories, the perfusion index and the relative diastolic notch height so we can model cardiovascular trends and provide early warning of hypotensive events.

The report is organized as follows:

1. Section 2: outlines the **methodology**, including STL operators, Stl Evaluator, extracting blood pressure cycle characteristics, MAP trajectory, perfusion index and relative diastolic notch height formulas.
2. Section 3: presents the experimental **results**.
3. Section 4: presents an **Analysis and Discussion** of the results.
4. Section 5: **concludes** by identifying the accomplishments, the framework’s limitations and directions for future work.

2 Methodology

2.1 Signal Temporal Logic

Signal Temporal Logic (STL) is a formal framework for specifying and reasoning about temporal properties of real-valued signals. By combining logical operators such as negation ($\neg$), conjunction ($\wedge$) and disjunction ($\vee$) with temporal operators including: G (always), F (Eventually), U (Until) and N (next), STL formulas can be constructed to evaluate whether a signal satisfies a desired temporal behavior or criterion over time [Maler and Nickovic, 2004]. An example of an STL formula:

$$\varphi_2 := G(x[t] > \pi \rightarrow F_{[0 \ \tau_1]}(G_{[0 \ \tau_2]} x[t] < \pi))$$

Which states that at all times, if the signal $x(t)$ rises above the threshold π , in the future after τ_1 seconds, the signal must always come below π for a time period of τ_2 seconds [Donze, 2013]. Formulas such as this can be applied to a signal over a

sliding window of time to determine exactly what point the formula is satisfied or not satisfied [Bakhirkin and Basset, 2019].

2.2 Stl Evaluator

Stl Evaluator (StlEval) is a tool developed by VERIMAG which extends the STL logic and analyzes STL formulas over continuous-time input signals [Bakhikin, 2019]. The tool processes input signals represented as time-series data from a CSV file where every line in the file corresponds to a timestamp and a corresponding signal value. StlEval then produces either a boolean or real-valued output with precise timestamps of the intervals which the signal satisfies or violates the specific formula. The standard StlEval framework requires static, hardcoded bounds for its temporal operators, thus it does not account for the variability in heart rates of different patients. To resolve this, an algorithm was developed that scales all temporal sliding evaluation windows based on the patients median heartbeat interval.

The methodology for extracting ABP cycle characteristics relies on two primary temporal operators. The $On()$ operator which evaluates an aggregate expression over a sliding window of time surrounding the current time and returns the signal value that satisfies the aggregate criteria within that time frame. It is used to return local values such as maxima, minima or averages in a signal. To isolate distinct features, the scalars of the search bounds for this operator were set to $\pm 40\%$ of the heartbeat interval for systolic and diastolic extrema and $\pm 10\%$ for identifying the diastolic notch. The $Until()$ operator specifies that one condition or event must remain valid until another condition becomes true within a specified time interval. It is used to define the maximum amount of time the logic will wait to find the next heartbeat. To prevent the search window from overlapping with the subsequent heartbeat and maintain that the evaluation logic remains confined to the current cardiac cycle, the upper bound of the $until()$ operator has been scaled to 90% of the median heart interval.

2.3 Extracting Blood Pressure Cycle Characteristics

Arterial Blood Pressure cycle

The arterial blood pressure (ABP) waveform consists of alternating systolic and diastolic phases (Figure 1). Three characteristic points are of particular interest in this study: the diastolic minimum, corresponding to the lowest pressure between heartbeats; the systolic peak, corresponding to the maximum pressure during ventricular ejection; and the diastolic notch, a local depression associated with aortic valve closure. These features serve as the primary landmarks used for waveform characterization and hypotension prediction.

Median Heartbeat Interval

The Median Heartbeat Interval (MHI) is the time it takes for one full heartbeat cycle. This variable can be computed as follows:

$$MHI = 60 / \text{currentheartrate}$$

Diastolic Blood Pressure

The Diastolic or ‘resting’ blood pressure occurring between successive heartbeats can be characterized with the following

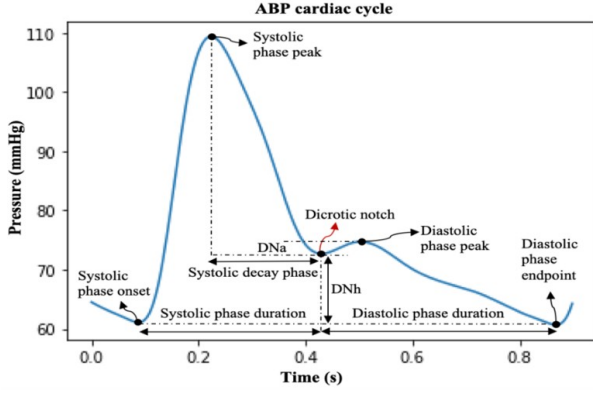


Figure 1: Arterial Blood Pressure Cycle [Pal *et al.*, 2024]

expression:

$$\theta_1 = \text{Min}(S(c_1, c_2)) \leq S(t)$$

Where $S(t)$ denotes the ABP signal at time (t) . c_1 and c_2 define the temporal bounds of a sliding observation window. This formula identifies points where the current signal value correspond to the minimum value within the surrounding temporal neighborhood and captures the minimum value of the ABP waveform. In the semantics of the StEval framework, the corresponding query to extract the value of θ_1 at the lowest point of the wave form is expressed as:

$$(\text{Until}(0.9(MHI)) \ 0 \ (\text{Get } x0) \\ (\geq (\text{On}(-0.4(MHI) \ 0.4(MHI))(\text{Min } S(t)))S(t)))$$

This query evaluates the blood pressure signal over a sliding temporal window of $(\pm 0.4 * MHI)$ seconds and identifies points where the current signal value equals the local minima. The (Get x0) operator is used to return the numerical value of the signal at the detected diastolic point.

Systolic Blood Pressure

The Systolic or ‘active’ blood pressure of each successive heartbeat can be characterized with the following expression:

$$\theta_2 = \text{Max}(S(c_1, c_2)) \geq S(t)$$

Where again, $S(t)$ denotes the ABP signal at time (t) and c_1 and c_2 define the temporal bounds of a sliding observation window. This formula now identifies points where the current signal value correspond to the maximum value within the surrounding temporal neighborhood and captures the maximum value of the ABP waveform. In the semantics of the StEval framework, the corresponding query to extract the value of θ_2 at the highest point of the wave form is expressed as:

$$(\text{Until}(0.9(MHI)) \ 0 \ (\text{Get } x0) \\ (\leq (\text{On}(-0.4(MHI) \ 0.4(MHI))(\text{Max } S(t)))S(t)))$$

This query evaluates the blood pressure signal over a sliding temporal window of $(\pm 0.4 * MHI)$ seconds and identifies points where the current signal value equals the local maximum. The (Get x0) operator returns the numerical value of the signal at the detected systolic point.

The Dicrotic Notch

To identify the dicrotic notch, a formula similar to θ_1 is used to detect the local minima. However, the temporal observation window is reduced to a smaller window denoted by s_1 and s_2 in order to capture all local depressions in the ABP waveform. To distinguish the dicrotic notch from the primary diastolic minima, an additional constraint is included to exclude points satisfying the condition for θ_1 the global minimum. The resulting formula is given by:

$$\theta_3 = \text{Min}(S(s_1, s_2)) \leq S(t) \wedge \neg \theta_1$$

Where $S(t)$ denotes the ABP signal at time (t) . s_1 and s_2 define a smaller temporal bounds of a sliding observation window while excluding the global minimum θ_1 , previously identified within the cardiac cycle. In the semantics the STL framework, the corresponding STL query to extract the value of θ_3 at the dicrotic notch is expressed as:

$$(\text{Until}(0.9(MHI)) \ 0 \ (\text{Get } x0) \\ (\text{and } (\geq (\text{On}(-0.1(MHI) \ 0.1(MHI))(\text{Min } S(t)))S(t)) \\ (\text{not } (\geq (\text{On}(-0.4(MHI) \ 0.4(MHI))(\text{Min } S(t)))S(t))))))$$

This query evaluates the waveform using a narrow temporal window of $(\pm 0.1 * MHI)$ seconds to identify all local minima corresponding to potential dicrotic notches and excludes points identified as the primary diastolic minima within the larger $(\pm 0.4 * MHI)$ window associated with θ_1 . The (Get x0) operator is used to return the numerical value of the signal at the detected dicrotic notch location.

2.4 Perfusion Index and Relative Dicrotic Notch Height

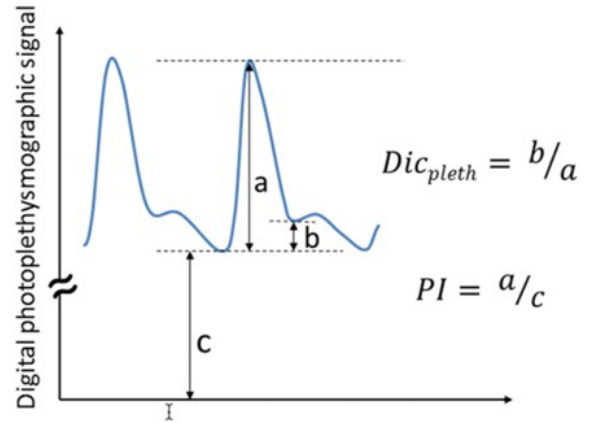


Figure 2: Dic_{pleth} and Perfusion Index [Coutrot *et al.*, 2019]

The Perfusion Index (PI) is a numerical measure of the strength of the pulse. The lower the number, the worse the bloodflow. It is calculated as a ratio between the systolic peak and the diastolic trough as shown in figure 2.

The Relative Dicrotic Notch Height (Dic_{pleth}) tells us about the systemic vascular resistance and arterial stiffness [Abushouk *et al.*,]. The Dic_{pleth} is defined as a ratio between the height of the dicrotic notch to the height of the systolic peak as shown in figure 2.

Predicting Hypotension using the Dicrotic Notch

According to Coutrot et al [2019], an increase in PI of over 51% from the resting baseline before induction and a reduction of Dic_{pleth} of 19% from a resting baseline before induction together can show indications of hypotension. This study could not obtain a resting baseline for PI or Dic_{pleth} so it used a calculated average of the first 100 registered heartbeats. Computing the deltas is as follows:

$$\Delta DIC_{pleth} = \frac{DIC_{pleth:hb} - DIC_{pleth:bl}}{DIC_{pleth:bl}} * 100$$

$$\Delta PI = \frac{PI_{hb} - PI_{bl}}{PI_{bl}} * 100$$

2.5 Mean Arterial Pressure (MAP)

The Mean Arterial Blood Pressure (MAP) is the average arterial pressure throughout a single cardiac cycle, encompassing both systolic and diastolic phases [DeMers and Wachs, 2023]. A commonly used approximation for computing the MAP is expressed by the following formula:

$$MAP := diastolic + \frac{(systolic - diastolic)}{3}$$

Predicting Hypotension using the MAP Trajectory

By computing the average MAP across n consecutive heartbeats, a vector is constructed which can be used to evaluate the current trajectory of arterial pressure over time. This trajectory can then be analyzed to estimate the time remaining until the MAP crosses a pressure threshold if the same trajectory persists. Based on this trajectory estimation, an early warning signal can be generated if/when the predicted time to hypotension falls below a predefined time threshold. Assuming hypotension occurs at 65 mmHg, the time to hypotension (TTH) can be computed with the following equation:

$$TTH = \Delta t * (p_c - 65) / \Delta p$$

Where Δp is the change in MAP from p_p (previous pressure) to p_c (current pressure) over the observed interval between t_p (previous time) and t_c (current time) represented by Δt (see Figure 3).

Data Requirements

The data used for this study was extracted from VitalDB; a high fidelity, vital signs database from the Department of Anesthesiology and Pain Medicine and the Seoul National University College of Medicine in Seoul, Korea. The data for this research project required high definition ABP of at least 100HZ frequency (data measures at 100 samples per second). It also needed cases where the blood pressure fell under 65 mmHg for at least 60 seconds. Furthermore, to compute the relative dicrotic notch height and perfusion index, it required cases where the dicrotic notch was well defined and followed the order of diastolic minima, systolic peak and dicrotic notch. Due to the StlEval Framework, all data must start with time at 0. A time = 0 and ABP = 0 (0,0) was added as the first parameter of all the cases used.

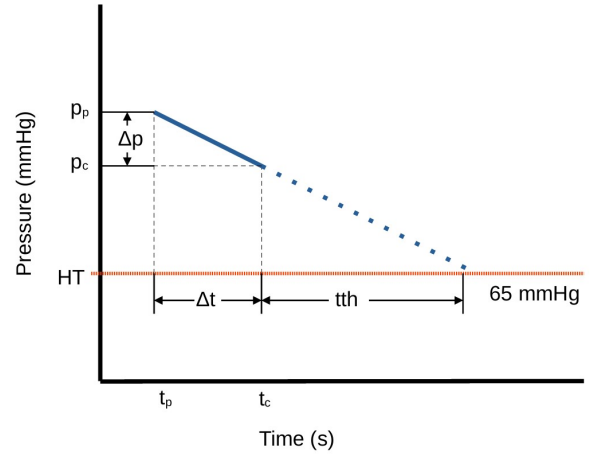


Figure 3: Trajectory Calculation for Time To Hypotension

3 Results

Figures 4, 5 and 6 illustrate the extraction of waveform characteristics using StlEval from representative VitalDB cases, including well-defined, missing, and multiple dicrotic notch morphologies. Figure 7 demonstrates hypotension prediction using MAP trajectory analysis. Figure 8 demonstrates both MAP trajectory and dicrotic-notch-derived metrics with ΔPI and ΔDic_{pleth} .

4 Analysis and Discussion

Extracting features of the blood pressure waveform

By using STL formulas within the StlEval framework, this project was able to successfully characterize and extract the precise timing and values of key features within real ABP waveforms. This includes: the diastolic minima (θ_1), the systolic peak (θ_2) and the dicrotic notch (θ_3) (see Figure 4) provided the input signal was well defined with a dicrotic notch.

Accurately characterizing the dicrotic notch (θ_3) proved challenging due to variability in waveform shape. In particular, certain blood pressure signals exhibited multiple dicrotic notches (see Figure 6), inverted or reverse dicrotic notch structures, or the complete absence of a clearly distinguishable dicrotic notch (see Figure 5). More research is required in order to positively identify the exact location of a dicrotic notch with these types of wave forms using STL and StlEval.

Predicting hypotension using the dicrotic notch

Given a detected diastolic minimum (θ_1), systolic peak (θ_2), and a defined dicrotic notch (θ_3), this study was unable to verify the ability to predict hypotension using the relative change in Perfusion Index (ΔPI) and Relative Dicrotic Notch Height (ΔDic_{pleth}) as proposed by Coutrot et al [2019]. In our dataset, although the computed ΔDic_{pleth} values decreased along with the MAP, they persistently registered below the previously established -19% threshold regardless of the underlying blood pressure. Furthermore, ΔPI diverged from the expectations given by the literature, decreasing rather than increasing as the MAP declined (See figure 8). Consequently, the parameters lacked the discriminative power necessary to serve as a reliable early warning indicator.

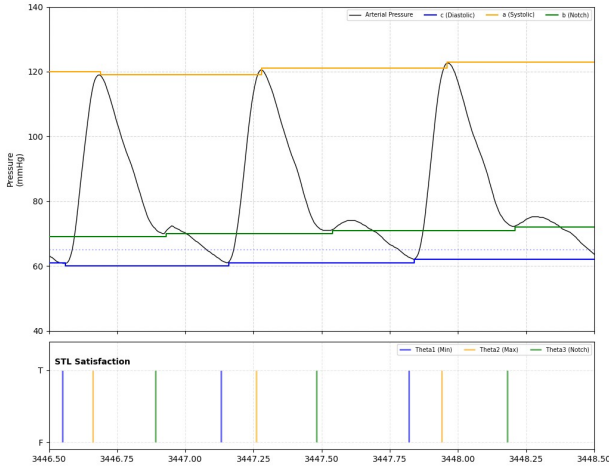


Figure 4: Three ABP Cycles; Full dicrotic notch [VitalDB, Case-6280]

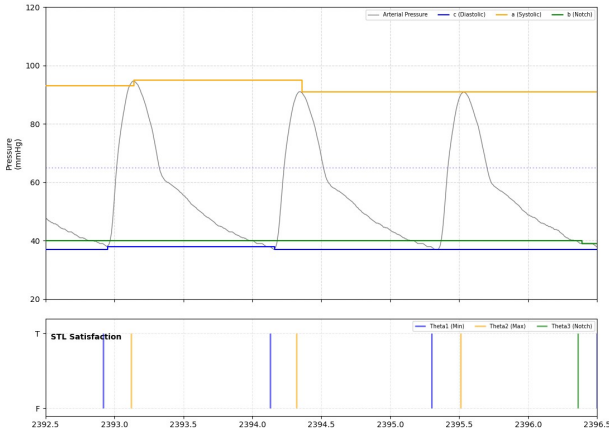


Figure 5: Three ABP Cycles; Missing dicrotic notch (θ_3). [VitalDB, Case-6102]



Figure 6: Three ABP Cycles; Multiple dicrotic notches (θ_3). [VitalDB, Case-166]

Predicting hypotension using the MAP trajectory

Using the detected diastolic minima (θ_1) and systolic peaks (θ_2), this project was able to compute the Mean Arterial Pressure (MAP), average the MAP and build a trajectory over time, then generate warnings when the estimated time to hypotension falls below a predefined threshold. In figure 7, when the trajectory of the average MAP computed over 20 heartbeats computes a time to hypotension of less than one minute, the warning will trigger. In the evaluated cases, the method successfully identified all observed hypotensive events. However, this method is also sensitive to noise. Very sharp decreases in the averaged MAP signal will cause the warning to trigger. Expanding the MAP average window can reduce false positives by smoothing signal noise. However, because the system relies on retrospective data, it increases the latency and reduces the system's warning time. Increasing the Time-To-Hypotension (TTH) threshold extends the predictive warning window to create hypotension warnings inherently increases the rate of false positive alerts.

5 Conclusion

In conclusion, this research demonstrates the efficacy of utilizing Signal Temporal Logic to evaluate the Arterial Blood Pressure waveform. By utilizing the StEval tool and developing scaled, sliding temporal evaluation windows based on a patient's heart rate, STL formulas were successfully applied to identify physiological features of the arterial blood pressure cycle including the diastolic minima, systolic maxima and the dicrotic notch. The extracted waveform characteristics were used to compute the average Mean Arterial Pressure trajectories and create an early warning of hypotension 1 minute before an onset. The study also explored the use of the dicrotic notch as a predictor of hypotension. Although preliminary studies indicate the potential value, variability in the notch morphology including: absent, multiple or inverted notches; limited its reliability within the current framework. Further investigation is required to determine if more sophisticated techniques or context can improve its utility.

Future work in this field should focus on 1): refining STL formulas for more robust waveform characterization, particularly identifying the dicrotic notch accurately, given the alternate conditions described in this paper. 2): Using the dicrotic notch to validate the studies by [Coutrot *et al.*, 2019] and [Aubouin *et al.*, 2025] to calculate the change in Relative Dicotic Notch Height from a resting baseline. 3): Evaluating the approach across larger and more diverse populations. 4): Converting the coding algorithm from using the cases from VitalDB to using a real-time signal directly from a patient.

Ultimately, this work demonstrates that dynamically scaled STL formulas provide a promising foundation for real-time monitoring and prediction of intraoperative hypotension, with the potential to support earlier clinical intervention and improve patient outcomes.

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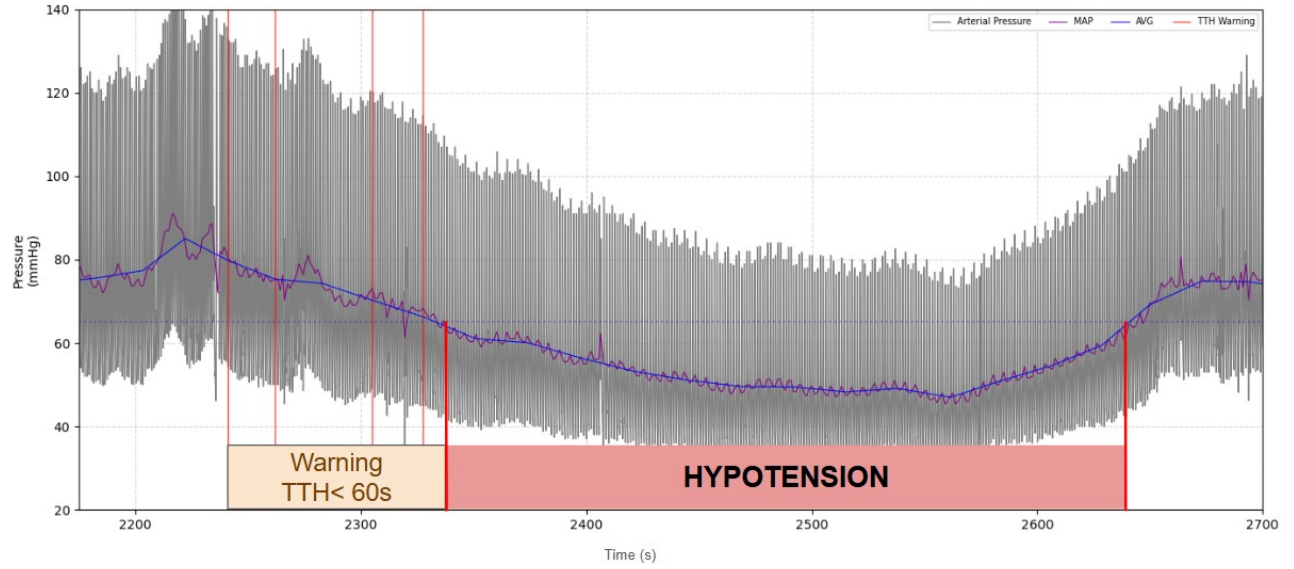


Figure 7: Time To Hypotension Warning using Avg MAP trajectory calculation. [VitalDB, Case-6102]

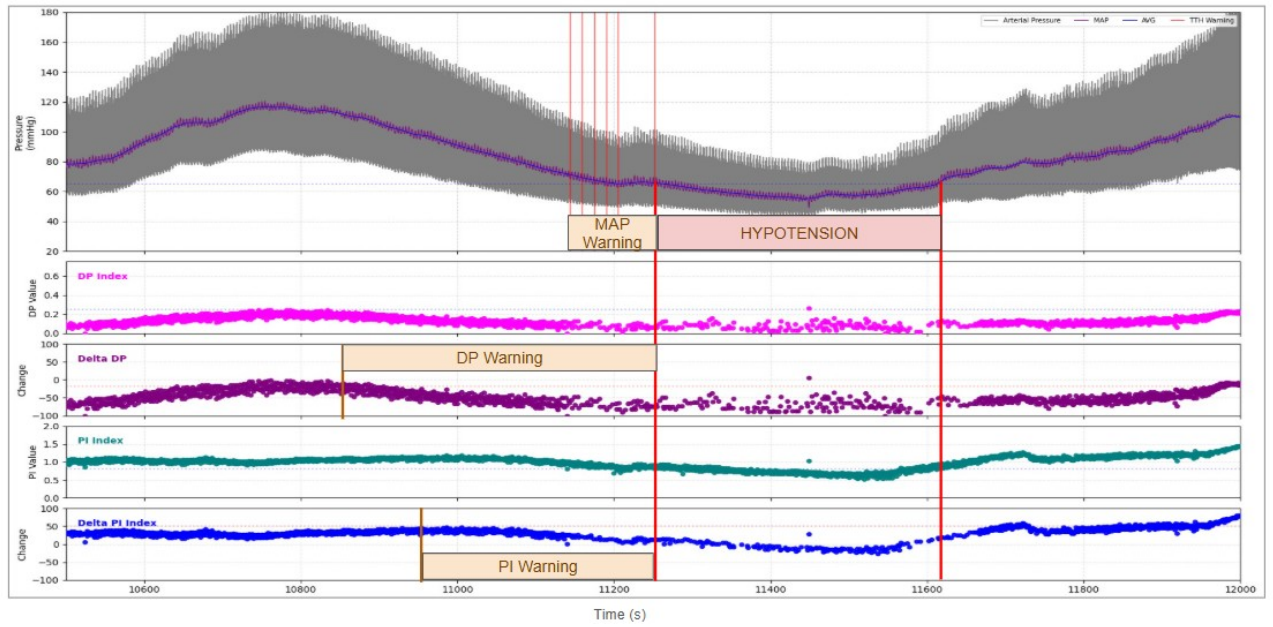


Figure 8: Time To Hypotension Warning using Avg MAP trajectory and Dicrotic Notch. Baseline $Di_{c_{pleth}} = 0.25$, Baseline PI = 0.8 [VitalDB, Case-10]

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